ASSIGNMENT

SOIL CLASSIFICATION

Subject: Soil Mechanics

Topic: Soil Classification

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1. Introduction

Soil is a natural material found on the Earth's surface. It consists of mineral particles, organic matter, water, and air. Soil is important in civil engineering because it supports buildings, bridges, roads, dams, and other structures.

Different soils have different properties. Some soils are strong and suitable for construction, while others may have low strength or undergo excessive settlement.

Soil classification is the process of grouping soils according to their physical properties, particle sizes, and engineering behavior. It helps engineers understand soil properties and select suitable soils for construction projects.

2. Objectives of Soil Classification

• To identify different types of soil.

• To understand the engineering properties of soil.

• To determine the suitability of soil for construction.

• To estimate the strength and compressibility of soil.

• To help in the design of foundations, roads, and earth structures.

• To provide a common system for describing soil.

3. Types of Soil

3.1 Gravel

Gravel consists of relatively large soil particles. It allows water to pass through it easily and generally has good drainage properties.

Characteristics: Large particles; high permeability; good drainage; generally good load-bearing capacity when properly compacted; and low compressibility in many well-graded gravel deposits.

Uses: Road construction, foundation beds, drainage layers, and concrete aggregate.

3.2 Sand

Sand consists of medium-sized soil particles. It is usually granular and can be identified by its gritty feel.

Characteristics: Particles are visible to the naked eye; good drainage; generally low plasticity; strength depends on density and particle arrangement; loose sand may settle under loading.

Uses: Concrete and mortar, road construction, bedding material, and foundation construction when suitable.

3.3 Silt

Silt consists of fine soil particles smaller than sand. It may feel smooth or flour-like when dry.

Characteristics: Fine particles; low plasticity compared with clay; may lose strength when wet; can be easily eroded; drainage is slower than in clean sand.

Uses: Earth filling and embankments when suitable and properly tested and compacted; agricultural soil when mixed with other components.

3.4 Clay

Clay consists of very fine particles and contains minerals that influence its plasticity and behavior with water.

Characteristics: Very fine particles; usually low permeability; becomes plastic when wet; may shrink when drying and swell when wet; can undergo significant settlement under loading.

Uses: Bricks and ceramics, impermeable barriers and liners when properly designed, and embankments when suitable. Clay soils require careful investigation before supporting foundations.

3.5 Organic Soil

Organic soil contains significant amounts of decomposed plant or animal material.

Characteristics: Usually dark in color; may contain plant remains; often highly compressible; can undergo considerable settlement; properties may change as organic material decomposes.

Uses and limitations: Organic soil may be useful for agriculture. Highly organic soil is generally unsuitable for supporting foundations without special investigation and treatment.

4. Main Soil Classification Systems

Two widely used soil classification systems in civil engineering are the Unified Soil Classification System (USCS) and the AASHTO Soil Classification System.

4.1 Unified Soil Classification System (USCS)

The USCS classifies soils according to particle-size distribution, the amount of fine particles, and the plasticity characteristics of the fines. It is widely used in geotechnical engineering.

Soil Group	Symbol	Description
Well-graded gravel	GW	Gravel with a suitable range of particle sizes
Poorly graded gravel	GP	Gravel with a limited range of particle sizes
Silty gravel	GM	Gravel containing silty fines
Clayey gravel	GC	Gravel containing clayey fines
Well-graded sand	SW	Sand with a suitable range of particle sizes
Poorly graded sand	SP	Sand with a limited range of particle sizes
Silty sand	SM	Sand containing silty fines
Clayey sand	SC	Sand containing clayey fines
Silt	ML or MH	Inorganic silt of low or high plasticity
Clay	CL or CH	Inorganic clay of low or high plasticity
Organic soil	OL or OH	Organic silt or clay of low or high plasticity
Peat	PT	Highly organic soil

Meaning of common USCS symbols: G = Gravel; S = Sand; M = Silt; C = Clay; W = Well graded; P = Poorly graded; L = Low plasticity; H = High plasticity; O = Organic soil; PT = Peat.

For example, CL represents inorganic clay of low plasticity, while CH represents inorganic clay of high plasticity.

Note: Exact classification depends on laboratory test results and the applicable classification criteria.

4.2 AASHTO Soil Classification System

The AASHTO system is commonly used in highway and pavement engineering. It classifies soils according to their suitability as subgrade material beneath roads.

This system considers particle-size distribution, percentage passing the No. 200 sieve, liquid limit, and plasticity index.

Group	General Description
A-1	Gravelly and sandy soils with favorable subgrade properties
A-2	Granular soils containing significant amounts of silt or clay
A-3	Fine sand
A-4	Silty soils with relatively low liquid limits

Group	General Description
A-5	Silty soils with relatively high liquid limits
A-6	Clayey soils with relatively low liquid limits
A-7	Clayey soils with relatively high liquid limits

The A-7 group can be further divided into A-7-5 and A-7-6 according to the relationship between the plasticity index and liquid limit. In general, A-1 soils tend to provide better road-subgrade performance than A-6 or A-7 soils, but actual performance also depends on moisture, compaction, and drainage.

5. Classification Based on Particle Size

Soils can also be classified according to the size of their particles. The following boundaries are commonly used in introductory soil mechanics; exact definitions may vary with the standard.

Soil Type	Approximate Particle Size
Gravel	4.75 mm to 75 mm
Sand	0.075 mm to 4.75 mm
Silt	Approximately 0.002 mm to 0.075 mm
Clay	Smaller than approximately 0.002 mm

Important: Clay is not identified by particle size alone. Its plasticity and behavior with water are also important.

6. Important Properties Used in Soil Classification

6.1 Grain-Size Distribution

Grain-size distribution describes the different sizes of particles present in a soil sample. It is determined using sieve analysis for coarse-grained soils and hydrometer analysis for fine-grained soils. These tests help determine the percentages of gravel, sand, silt, and clay-sized particles.

6.2 Atterberg Limits

Atterberg limits describe the consistency of fine-grained soils at different moisture contents.

• **Liquid Limit (LL):** The moisture content at which soil changes from a plastic state to a liquid-like state under specified test conditions.

• **Plastic Limit (PL):** The moisture content at which soil begins to change from a semi-solid state to a plastic state.

• **Plasticity Index (PI):** The range of moisture content over which the soil behaves plastically.

Formula: PI = LL – PL

Example: If LL = 45% and PL = 25%, then PI = 45 – 25 = **20%**.

6.3 Moisture Content

Moisture content is the amount of water present in soil compared with the mass of dry soil. It affects soil strength, compaction, and consistency.

6.4 Permeability

Permeability is the ability of soil to allow water to pass through it. Gravel and clean sand generally have higher permeability than clay.

6.5 Plasticity

Plasticity is the ability of fine-grained soil to change shape without cracking or breaking and to retain that shape after the applied force is removed. Clay generally exhibits more plasticity than silt.

7. Laboratory Tests Used for Soil Classification

7.1 Sieve Analysis

Sieve analysis determines the particle-size distribution of coarse-grained soil.

Procedure: (1) Take a representative soil sample. (2) Dry the sample. (3) Arrange sieves in decreasing order of opening size. (4) Place soil on the top sieve. (5) Shake the sieve stack. (6) Weigh soil retained on each sieve. (7) Calculate the percentage retained and percentage passing.

7.2 Hydrometer Analysis

Hydrometer analysis estimates the particle-size distribution of fine-grained soil particles suspended in water. It is based on the rate at which particles settle through water.

7.3 Liquid Limit Test

This test determines the liquid limit of fine-grained soil. It helps evaluate soil consistency and is used in classification systems.

7.4 Plastic Limit Test

This test determines the moisture content at which soil begins to lose its plastic behavior during the specified rolling procedure.

7.5 Natural Moisture Content Test

This test determines the water content of a soil sample and helps engineers understand the soil's condition at the time of sampling.

8. Importance of Soil Classification in Civil Engineering

• **Foundation design:** Helps engineers select suitable foundation types and investigate soil bearing capacity.

• **Road construction:** Helps evaluate soil suitability for road subgrades and pavement layers.

• **Building construction:** Helps identify soils that may cause settlement or swelling.

• **Earth dams:** Helps select suitable materials for different zones of an earth dam.

• **Drainage design:** Helps engineers understand how water moves through soil.

• **Slope stability:** Provides information useful for assessing the behavior of soil slopes.

• **Quality control:** Allows engineers to compare soil samples and check construction materials.

• **Compaction:** Helps in selecting suitable compaction methods and evaluating soil behavior.

9. Difference Between USCS and AASHTO Systems

Basis	USCS	AASHTO
Full form	Unified Soil Classification System	AASHTO soil classification system
Main application	General geotechnical engineering	Highway and pavement engineering
Classification	Symbols such as GW, SP, CL, and CH	Groups such as A-1, A-2, A-4, and A-7
Important properties	Grain-size distribution and plasticity	Grain-size distribution, liquid limit, and plasticity index
Main purpose	Describes soil type and engineering characteristics	Evaluates soil for highway subgrade applications

10. Conclusion

Soil classification is an essential part of civil engineering. It helps engineers identify different types of soil and understand their engineering behavior.

The major soil types include gravel, sand, silt, clay, and organic soil. The Unified Soil Classification System (USCS) and the AASHTO system are widely used for classifying soils.

Laboratory tests such as sieve analysis, hydrometer analysis, liquid limit testing, and plastic limit testing provide the information needed for classification.

Proper soil classification helps engineers make better decisions about foundations, roads, embankments, and other civil engineering structures. However, classification alone is not enough to determine whether soil is safe for a particular structure. Additional tests and site investigations are often required.